

L02 Image Processing Fundamentals

Technical Understanding

The most surprising discovery from this lab was how directly digital images map to numbers. It reminded me of a funny elementary school memory where we would open image files in TextEdit and corrupt the data and “glitch” the image and call ourselves hackers. At the time, it felt like messing with the hidden code, but this lab helped me understand why those changes worked: the visible picture depends on structured numeric data. A color image looks like a normal picture when displayed on screen, but underneath it is a three-dimensional matrix made from height, width, and color channels. Each visible change in color, brightness, or contrast comes from changing numeric pixel values. I have used image editing tools before, but this made the process feel less abstract because I could see how the visual result came from controlled mathematical changes.

The mathematical operations we implemented related to visual effects in a very direct way. Brightness was changed by adding values to pixels, contrast was changed by spreading pixel values farther from the midpoint, and convolution filters changed the image by using neighboring pixels. Blur reduced fine detail by averaging nearby values, while edge detection and sharpening emphasized changes between neighboring intensities. The technique that was most difficult to comprehend for me was convolution. I understood that a kernel is a small grid of numbers used as a filter, but it was harder to predict what the kernel would do until I saw the output image. Brightness and contrast felt more direct, while convolution made more sense once I could compare the blur, sharpening, edge detection, and emboss results visually.

Connections and Applications

This lab connected to the Nano Banana demonstration because both traditional image processing and AI image tools work with visual features such as edges, textures, color, contrast, and spatial relationships. The difference is that traditional processing uses filters and formulas chosen by the programmer, while AI tools learn more complex combinations from data. Understanding the mathematical foundation makes AI image tools seem less like magic and more like advanced systems built on many image operations working together.

The techniques from this lab have many practical applications. Histogram enhancement and CLAHE can improve low-contrast images, blur can reduce noise,

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sharpening can restore edge detail, and geometric transformations can rotate or correct image orientation. These methods could be used for all sorts of post-processing improvements. Most photo editing software contain “auto” enhancement techniques - and I can see how looking at the data from an image it could easily be determined what techniques could improve the image to a more useful state. This could be implemented for smartphone camera adjustments, document scanning, OCR cleanup, and photo restoration. In a future project, I could see how combining traditional and AI approaches by cleaning up an image first with contrast enhancement, denoising, or sharpening techniques before sending it to the machine learning model would provide better, more uniform inputs so that the AI model would train on and produce more accurate and polished results.

Personal Reflection

The aspect of image processing that interests me most is how preprocessing choices affect computer vision accuracy. The cleanup pipeline experiment was especially interesting because it combined image transformation into a practical workflow instead of just creating a decorative effect. This reminds me of how photographers are able to sell photo editing presets. This concept previously mystified me, however, it is clear that specific combinations of contrast, blur, sharpening, and color adjustments create a recognizable and repeatable style. In this assignment we saw that CLAHE improved local contrast, blur reduced small artifacts, and sharpening restored edge detail. That made me think more about how image processing can be used as a bulk preparation step before analysis, not only as a final editing step.

This lab changed my understanding of digital photography and image editing by showing that many familiar editing tools are based on simple numerical operations. I had thought about brightness, contrast, blur, and sharpening mainly as visual sliders, but now I understand that they are built from specific changes to pixel values and neighborhoods. My lingering questions are about how image levels and curves work mathematically, how much preprocessing is ideal before using an AI model, and how to decide when traditional image processing is better than an AI-powered approach.